The Journal of Chemical Physics

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W. Küchle, M. Dolg, H. Stoll, and H. Preuss

Citation: J. Chem. Phys. 100, 7535 (1994); doi: 10.1063/1.466847
View online: http://dx.doi.org/10.1063/1.466847
View Table of Contents: http://jcp.aip.org/resource/1/JCPSA6/v100/i10
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Energy-adjusted pseudopotentials for the actinides. Parameter sets and test calculations for thorium and thorium monoxide

W. Küchle, M. Dolg,[a] H. Stoll, and H. Preuss
Institut für Theoretische Chemie, Universität Stuttgart, D-70550 Stuttgart, Germany

(Received 20 December 1993; accepted 7 February 1994)

We present nonrelativistic and quasirelativistic energy-adjusted pseudopotentials, the latter augmented by spin–orbit operators, as well as optimized $(12s11p10d8f)/[8s7p6d4f]$-Gaussian-type orbitals (GTO) valence basis sets for the actinide elements actinium through lawrencium. Atomic excitation and ionization energies obtained by the use of these pseudopotentials and basis sets in self-consistent field (SCF) calculations differ by less than 0.2 eV from corresponding finite-difference all-electron results. Large-scale multiconfiguration self-consistent field (MCSCF), multireference configuration interaction (MRCI), and multireference averaged coupled-pair functional (MRACPF) calculations for thorium and thorium monoxide yield results in satisfactory agreement with available experimental data. Preliminary results from spin–orbit configuration interaction calculations for the low-lying electronic states of thorium monoxide are also reported.

I. INTRODUCTION

Pseudopotential methods have proven to be very useful for quantum theoretical investigations of chemical systems containing heavy elements.[1,2] While for reasonable choices of the core the results of pseudopotential calculations usually are in rather good agreement with all-electron treatments (cf. e.g. Ref. 3), the computational effort is considerably reduced due to a smaller number of basis functions needed for the treatment of the valence-electron system. Clearly, this advantage is only present when the innermost nodes are removed from the pseudo-orbitals, as is done for most sets of pseudopotentials or effective core potentials published in literature. Furthermore, when adjusting the pseudopotentials to spin–orbit averaged (quasi-)relativistic atomic data, they provide a reliable tool for easily covering the major relativistic effects within formally nonrelativistic self-consistent field (SCF) and subsequent configuration interaction (CI) calculations.

Quasirelativistic pseudopotentials for some or all of the first actinide elements Ac through Pu have been published by various authors,[4–9] however, to our knowledge no complete set of pseudopotentials is available for the whole $5f$ series Ac through Lr. Surprisingly there seem to be no nonrelativistic pseudopotentials at all for those elements, although they are needed for the proper discussion of relativistic effects in actinide compounds. Therefore we present quasirelativistic as well as nonrelativistic pseudopotentials together with corresponding optimized valence basis sets for all of the actinide elements.

For transition elements with a partially occupied $(n-1)d$ subshell $(n=4-6)$ frozen-core calculations (as well as pseudopotential calculations) on excitation and ionization energies are known to show significant errors, if the $(n-1)s$ and $(n-1)p$ subshells are included into the core. This is due to non-negligible changes of interaction between these subshells and the $(n-1)d$ valence subshell when changing the occupation number of the latter. Therefore accurate calculations require the explicit treatment of the complete $(n-1)$ shell together with the ns and np valence orbitals, considerably raising the number of valence electrons and the concomitant computational effort. The f elements are even more complicated. In most ground states and low-lying excited states the $(n-1)d$ $(n=6,7)$ as well as the $(n-2)f$ subshell are partially filled. Inclusion of s, p, and d orbitals for $n-2$ and the s and p orbitals for $n-1$ into the core leads to sizable frozen-core errors depending on the d *and* f occupation numbers. For the rare earth elements, La through Lu, it was possible to generate pseudopotentials with the open $4f$ subshell included into the core.[10] Especially for the first elements of the actinide series this procedure is probably not suitable. The rather diffuse $5f$ orbitals significantly participate in chemical bonding and therefore have to be treated as valence orbitals. Consequently, in analogy to the alternate choice of the core for the lanthanides,[11] we decided to generate pseudopotentials with a small [Kr] $4d^{10}$ $4f^{14}$ core. Thus, the emphasis of this approach is on accuracy and less on computational savings. Preliminary results of complete active space self-consistent field (CASSCF), multireference configuration interaction (MRCI), and multireference averaged coupled-pair functional (MRACPF) calculations are presented for low-lying electronic states of Th and Th^+ as well as for the $^1\Sigma^+$ ground state of ThO. The influence of spin–orbit coupling on the derived molecular constants of the low-lying electronic states of ThO is also discussed.

II. METHOD

In this work we use an atomic valence model Hamiltonian of the form

$$H=-\frac{1}{2}\sum_i \Delta_i+\sum_i U(r_i)+\sum_{i<j} \frac{1}{r_{ij}} \qquad (1)$$

including a semilocal pseudopotential

[a]Ohio Supercomputer Center visitor 1991 and 1993.

$$U(r_i)=-\frac{Q}{r_i}+\sum_{l=0}^{4} A_l e^{-\alpha_l r_i^2} P_l. \tag{2}$$

In these equations, Q denotes the charge of the core, i and j represent electron indices, A_l and α_l are adjustable parameters, and P_l is the projection operator onto the Hilbert subspace with angular symmetry l,

$$P_l=\sum_{m_l} |lm_l\rangle\langle lm_l|. \tag{3}$$

Our pseudopotentials are adjusted by a least-squares fit to the valence energies of 10 to 13 low-lying electronic states of the neutral atom and the singly positive ion (multielectron fit, MEFIT). The valence energies needed for the construction of the pseudopotentials were taken from finite-difference all-electron Hartree Fock (HF) calculations.[12,13,14] The reference data for the quasirelativistic pseudopotentials have been obtained using the formalism of Wood and Boring[15] by adding a mass-velocity term

$$h_{\mathrm{mv},i}=-\left(\frac{\alpha^2}{2}\right)[V_i(r)-\epsilon_i]^2 \tag{4}$$

(α is the fine structure constant) and a Darwin term

$$h_{D,i}=-\left(\frac{\alpha^2}{4}\right)\frac{dV_i(r)}{dr} B_i\left(\frac{d}{dr}-\frac{1}{r}\right) \tag{5}$$

with

$$B_i=\left\{1+\left(\frac{\alpha^2}{2}\right)[\epsilon_i-V_i(r)]\right\}^{-1} \tag{6}$$

to the nonrelativistic Hartree Fock operator $F_{\mathrm{nr},i}$ of the ith orbital,

$$[F_{\mathrm{nr},i}+h_{\mathrm{mv},i}+h_{D,i}]\phi_i=\epsilon_i\phi_i. \tag{7}$$

The Wood–Boring Eq. (7) may be derived from the Dirac equation for a single particle in a central field $V(r)$ by elimination of the small components followed by approximately averaging over spin–orbit coupling. This scheme is similar to the method of Cowan and Griffin,[16] where the Darwin term $h_{D,i}$ is applied for $l=0$ only, which is correct to $O(\alpha^2)$. Although the higher order contributions of this term are small for $l\geq 1$ we prefer to keep it, since it causes no problems in atomic calculations.

For an atom with nuclear charge Z the potential $V_i(r)$ used in Eqs. (4)–(6) is approximated in a local form by

$$V_i(r)=-\frac{Z}{r}+V_{C,i}(r)+V_{X,i}(r) \tag{8}$$

with the HF Coulomb potential $V_{C,i}$ and Slater's local approximation for the HF exchange potential $V_{X,i}$ (Ref. 17) (multiplied by a scaling factor 2/3), whereas in the Fock-operator of Eq. (7) the correct nonlocal HF potential is used. To avoid errors due to the incomplete cancellation of self-interaction in the Coulomb and exchange potential, we use a direct potential without self-interaction terms and apply the self-interaction correction according to Perdew and Zunger[18] for the local exchange potential.

The iterative optimization procedure[12] requires an initial guess for the pseudopotential parameters α_l and A_l. Reasonable values may be obtained for each value of the quantum number l by means of a single electron fit (SEFIT) using the valence energies of the one-valence electron ions $An^{(Q-1)+}$ $1s^2...4f^{14}\ nl^1\ {}^2L$ ($n=5-8$; $l=s,p,d,f,g$; $L=S,P,D,F,G$) as reference data. Alternatively the parameters of the neighboring atom in the Periodic Table may be used. The SEFIT-parameters for $l=4$ were not further optimized in a MEFIT, because calculations for valence states with occupied $5g$ orbitals did not converge, neither for the neutral atom nor for the singly positive ion.

For the use of our quasirelativistic pseudopotentials within a two component formalism, additional spin–orbit operators of the form[19]

$$U_{\mathrm{SO}}=\sum_{l=1}^{3}\frac{2\Delta U_l}{2l+1} P_l \mathbf{ls} P_l \tag{9}$$

have been generated. The difference ΔU_l of the radial parts of the quasirelativistic pseudopotentials for $j=l\pm\frac{1}{2}$, $U_{l,l+1/2}$ and $U_{l,l-1/2}$, may be written in terms of Gaussian functions

$$\Delta U_l=\Delta A_l e^{-\alpha_l r^2}. \tag{10}$$

The exponents α_l were set equal to the values of the corresponding pseudopotential parameters while the coefficients ΔA_l have been adjusted to reproduce the spin–orbit splittings obtained by first-order perturbation theory using the orbitals from a quasirelativistic all-electron calculation and a spin–orbit operator (taken from the Wood–Boring equation before averaging over j values) of the form

$$h_{\mathrm{SO},i}=-\left(\frac{\alpha^2}{4}\right)\frac{dV_i(r)}{dr} B_i \frac{\kappa+1}{r}, \tag{11}$$

where κ denotes a relativistic quantum number defined as

$$\kappa=\mp(j+\tfrac{1}{2})\quad \text{for } j=l\pm\tfrac{1}{2}. \tag{12}$$

$(12s11p10d8f)/[8s7p6d4f]$ Gaussian-type orbitals (GTO) valence basis sets have been created for application in SCF-LCAO calculations. Most of the parameters have been energy-optimized for the experimental ground state of each element. For actinium and thorium, both having ground states without occupied $5f$ orbitals, the f basis has been optimized for $7s^2 5f^1\ {}^2F$ (Ac) and $7s^2 6d^1 5f^1\ {}^3H$ (Th), respectively.

The parameter sets of our quasirelativistic pseudopotentials are listed in Table I. The parameters of the nonrelativistic pseudopotentials and the basis sets (both for nonrelativistic and quasirelativistic calculations) are available from the authors.[20]

III. RESULTS

In Table II a few results of SCF calculations on excitation and ionization energies with our pseudopotentials and basis sets are compared with finite-difference all-electron SCF values (E_{num}). The deviations of finite-difference pseudopotential results (ΔE_{pp})—these are the errors attributed solely to the pseudopotential—are of the order of 0.1 eV for low-lying electronic states. It has to be noted, that the

TABLE I. Pseudopotential parameters (in atomic units).

	l	Quasirelativistic PP		
		α_l	A_l	ΔA_l
Ac	0	14.651 879 51	476.146 921 92	
	1	9.180 346 00	185.895 520 68	46.373 521 28
	2	9.523 163 67	204.176 034 73	22.679 989 12
	3	7.594 209 62	100.818 254 74	6.288 418 50
	4	11.027 732 33	−51.615 241 81	
Th	0	14.776 424 13	455.055 967 63	
	1	9.083 184 34	177.691 790 70	39.577 965 28
	2	9.653 857 71	197.232 705 11	21.055 544 24
	3	7.380 765 89	82.908 727 23	4.816 749 92
	4	11.609 024 62	−55.126 419 06	
Pa	0	15.269 173 67	475.475 732 83	
	1	9.957 736 99	207.664 859 62	53.688 246 24
	2	8.943 141 26	156.674 119 88	14.349 838 62
	3	7.010 466 62	67.982 236 85	3.637 672 90
	4	12.201 056 01	−57.630 599 89	
U	0	16.414 038 69	536.516 627 78	
	1	9.060 556 06	169.544 924 65	28.721 672 43
	2	8.831 831 98	142.615 598 37	12.014 385 43
	3	7.018 516 29	60.393 076 02	3.100 576 99
	4	12.804 088 44	−60.129 989 58	
Np	0	16.771 947 70	541.538 538 51	
	1	9.783 552 77	191.084 571 37	38.681 131 70
	2	9.034 226 60	146.402 589 23	12.224 037 28
	3	7.121 848 27	60.394 333 14	3.044 412 72
	4	13.418 322 53	−62.626 185 08	
Pu	0	18.559 689 16	662.756 241 98	
	1	9.473 516 97	179.415 289 79	27.791 589 74
	2	9.030 690 65	139.705 912 17	10.954 120 86
	3	7.138 797 79	56.045 909 14	2.741 999 84
	4	14.043 996 55	−65.121 216 11	
Am	0	19.130 096 48	672.426 908 48	
	1	9.559 142 07	180.983 630 04	24.547 021 05
	2	9.161 266 17	139.037 563 89	10.636 493 29
	3	7.035 754 63	51.273 553 14	2.424 955 27
	4	14.681 332 89	−67.616 942 08	
Cm	0	19.250 346 34	648.910 581 92	
	1	9.745 660 41	185.646 637 94	22.907 447 64
	2	9.267 023 14	137.840 261 76	10.176 878 35
	3	7.063 714 12	48.180 996 36	2.192 324 40
	4	15.330 525 65	−70.114 975 50	
Bk	0	20.285 376 10	696.475 122 20	
	1	9.868 396 47	188.760 265 72	20.029 735 46
	2	8.916 557 40	124.935 835 85	8.168 696 53
	3	7.025 003 14	45.216 507 78	2.018 511 17
	4	15.991 805 28	−72.617 188 80	
Cf	0	22.271 632 01	823.746 576 43	
	1	8.800 799 15	173.070 439 52	4.771 075 28
	2	9.311 157 46	125.568 751 16	8.509 665 81
	3	6.629 577 84	34.952 596 23	1.512 229 76
	4	16.665 369 73	−75.125 254 93	
Es	0	25.239 434 42	1 109.052 533 71	
	1	8.574 504 40	175.318 777 24	2.762 618 65
	2	9.454 607 30	124.586 814 14	8.351 855 79
	3	6.273 390 51	29.551 927 08	1.242 568 20
	4	17.351 456 18	−77.641 074 78	
Fm	0	26.499 713 56	1 201.222 799 83	
	1	8.455 091 35	179.580 219 04	1.738 922 29
	2	9.526 110 52	122.515 303 86	7.831 731 19
	3	6.483 649 25	29.499 639 60	1.277 552 61
	4	18.050 160 51	−80.165 201 55	
Md	0	26.257 029 41	1 099.410 318 34	
	1	9.144 677 44	192.251 427 77	2.716 414 55
	2	9.769 849 78	129.191 583 91	8.413 027 13
	3	6.967 531 40	34.329 321 33	1.506 947 91
	4	18.761 791 72	−82.700 152 85	
No	0	26.788 774 34	1 099.412 551 14	
	1	9.671 941 89	203.319 049 71	3.396 936 18
	2	9.659 545 37	127.627 445 57	7.874 425 31
	3	6.604 736 47	31.661 862 79	1.327 454 17
	4	19.486 525 44	−85.247 367 41	
Lr	0	41.623 097 50	4 933.050 204 35	
	1	9.667 011 76	208.559 352 32	2.393 587 25
	2	8.455 596 01	105.298 950 96	4.402 988 02
	3	7.391 012 31	36.225 703 62	1.658 660 15
	4	20.224 533 42	−87.807 982 52	

quality of our pseudopotentials has been carefully tested by performing calculations for typically about 30 electronic states for each element. As an example the full data for thorium are shown in Table III. The extensive test showed, that the quality of the nonrelativistic pseudopotentials remains equally good throughout the whole series from Ac to Lr, while the quasirelativistic pseudopotentials point to some loss of accuracy in reproducing all-electron results for the heavier elements. We ascribe this effect to the choice of the pseudopotential core. The approximation of including relativistic effects implicitly into the pseudopotential and treating the valence electrons in a formally nonrelativistic manner is clearly getting worse with increasing atomic number as well as decreasing number of electrons incorporated into the core. Therefore the heavier $5f$ elements seem to be a critical case for quasirelativistic pseudopotentials; the small pseudopotential core necessary to avoid significant frozen-core errors is probably not large enough to fully account for relativistic effects in the valence shell. Since the $5f$ orbitals are getting more corelike with increasing atomic number, it may be possible to derive accurate pseudopotentials for fixed f occupation with the open $5f$ shell included into the core for the transamericium elements. Due to the larger core those potentials should not suffer from an insufficient description of relativity, however at the price of this type of pseudopotentials being only suitable for cases where the actinide atom keeps an integral $5f$ occupation number within a molecular environment.

Another error of about 0.1 eV has to be taken into account when proceeding from finite-difference to SCF-LCAO calculations using our basis sets. This is shown by the third and sixth column in Table II (ΔE_{bas}) where the deviations of SCF-LCAO results from finite-difference all-electron values

TABLE II. Excitation and ionization energies from nonrelativistic and quasirelativistic finite-difference all-electron calculations E_{num}. ΔE_{pp} denote the errors due to the pseudopotential in finite-difference valence-only calculations, while ΔE_{bas} refers to the deviations due to pseudopotential *and* basis set in valence-only LCAO calculations. All values are given in eV.

			Nonrelativistic			Quasirelativistic		
			E_{num}	ΔE_{pp}	ΔE_{bas}	E_{num}	ΔE_{pp}	ΔE_{bas}
Ac	s^2d^1	^2D	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^1	^2F	0.6051	0.0219	0.0346	3.4349	0.0002	0.0022
Ac^+	s^2	^1S	6.0567	−0.0046	−0.0165	4.2963	0.0002	−0.0176
Th	s^2d^2	^3F	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	$s^2d^1f^1$	^3H	3.3979	−0.0471	−0.0485	1.3136	0.0009	0.0017
Th^+	s^2d^1	^2D	7.6820	−0.0109	−0.0114	5.7222	0.0093	0.0120
Pa	$s^2d^1f^2$	^4K	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^3	^4I	−2.6703	−0.0358	−0.0318	2.3878	−0.0304	−0.0449
Pa^+	s^2f^2	^3H	6.9468	−0.0015	−0.0161	5.3296	0.0120	0.0310
U	$s^2d^1f^3$	^5L	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^4	^5I	−3.3650	−0.0143	−0.0122	1.9636	−0.0208	−0.0071
U^+	s^2f^3	^4I	7.1899	0.0068	0.0078	5.4535	0.0233	0.0165
Np	$s^2d^1f^4$	^6L	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^5	^6H	−4.0001	0.0100	0.0000	1.6031	0.0050	0.0078
Np^+	s^2f^4	^5I	7.4199	0.0285	0.0224	5.5791	0.0281	0.0319
Pu	$s^2d^1f^5$	^7K	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^6	^7F	−5.6953	−0.0019	−0.0136	0.3157	−0.0113	−0.0164
Pu^+	s^2f^5	^5H	7.5784	0.0297	0.0625	5.6328	−0.0256	−0.0218
Am	$s^2d^1f^6$	^8H	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^7	^8S	−8.0183	−0.0276	−0.0018	−1.5378	0.0021	−0.0105
Am^+	s^2f^6	^7F	7.5377	0.0660	0.0978	5.4832	0.0041	0.0162
Cm	$s^2d^1f^7$	^9D	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^7	^8S	−3.5658	−0.0054	−0.0098	2.6825	−0.0402	−0.0457
Cm^+	s^2f^7	^8S	7.1722	0.0377	0.0633	5.0341	0.0365	0.0408
Bk	$s^2d^1f^8$	^8H	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^9	^6H	−6.3808	−0.0245	−0.0302	0.3619	−0.0495	−0.0544
Bk^+	s^2f^8	^7F	6.7386	0.0325	0.0588	4.5453	0.0482	0.0425
Cf	$s^2d^1f^9$	^7K	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^{10}	^5I	−8.3832	−0.0519	−0.0441	−1.1899	0.0098	0.0144
Cf^+	s^2f^9	^6H	6.6429	0.0465	0.0701	4.3648	0.0473	0.0489
Es	$s^2d^1f^{10}$	^6L	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^{11}	^4I	−9.0216	−0.0512	−0.0479	−1.4831	−0.0001	0.0001
Es^+	s^2f^{10}	^5I	6.7086	0.0453	0.0697	4.3246	0.0544	0.0347
Fm	$s^2d^1f^{11}$	^5L	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^{12}	^3H	−9.5876	−0.0191	−0.0207	−1.6998	−0.1121	−0.0429
Fm^+	s^2f^{11}	^4I	6.8224	0.0333	0.0593	4.3243	−0.0280	−0.0422
Md	$s^2d^1f^{12}$	^4K	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	s^2f^{13}	^2F	−11.4451	0.0246	0.1449	3.0846	0.0694	−0.0093
Md^+	s^2f^{12}	^3H	6.8812	0.0139	0.0305	4.2736	0.0576	0.0155
No	s^2f^{14}	^1S	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	$s^2d^1f^{13}$	^2D	13.9980	0.0003	0.0270	5.0775	0.0003	−0.0301
No^+	s^1f^{14}	^2S	4.4749	0.0407	0.0389	5.4717	0.0372	−0.0207
Lr	$s^2d^1f^{14}$	^2D	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	$s^1d^2f^{14}$	^4F	−0.4411	−0.0423	−0.0337	2.3273	0.0118	0.0529
Lr^+	$s^1d^1f^{14}$	^3D	4.6168	0.0022	0.0040	5.9965	0.0624	0.0694

are listed. In some cases the errors due to pseudopotential and basis set cancel but clearly this is not expected to happen in each case. Therefore we expect our pseudopotentials and basis sets to reproduce finite-difference SCF values for excitation and ionization energies of reasonably low-lying electronic states with a deviation of less than 0.2 eV.

A. Atomic calculations on Th and Th^+

We performed correlated calculations for some low-lying electronic states of neutral thorium and the singly positive ion using the program system MOLPRO.[21–26] The basis set was augmented by two g functions with exponents of 1.938 and 0.651 which have been energy-optimized within a small configuration interaction calculation including all single and double excitations [CI(SD)] for the $7s^2\ 6d^1\ 5f^1\ ^3H$ state of thorium correlating only the four energetically highest valence electrons. In our subsequent calculations excitations from the $6p$ subshell have been taken into account, increasing the number of correlated electrons to 10 for the neutral atom. MRACPF (Ref. 26) calculations have been preceded by state-averaged CASSCF (Refs. 22,23) with $7s$, $5f$, $6d$, and $7p$ included into the active space. The CASSCF eigenvectors have been used for selecting the reference states for the MRACPF calculations. All configuration state functions with CI coefficients greater than a given threshold (0.09) have been incorporated into the reference space. For the two states Th $7s^2\ 6d^1\ 7p^1\ ^3F$ and Th $7s^2\ 6d^1\ 5f^1\ ^3H$ only $7s$, $6d$, and $7p$ or $7s$, $6d$, and $5f$ respectively have been included into the active space and the complete active space has been taken as reference for the subsequent MRACPF calculations. (Of course, we also performed corresponding calculations for the ground state to avoid errors of the excitation energies due to different active spaces.) We give the following reason for the special treatment of these states: both belong to a group of triplet states ($7s^2\ 6d^1\ 5f^1\ ^3H, G, F, D, P$; $7s^1\ 6d^2\ 5f^1\ ^3I,...$; $7s^2\ 6d^1\ 7p^1\ ^3F, P$; $7s^1\ 6d^2\ 7p^1\ ^3G,...$) strongly mixing with each other. Including $7s$, $5f$, $6d$, and $7p$ into the active space leads to a long CASSCF eigenvector with no obvious gap between the CI coefficients. Therefore, selecting reference configurations according to the magnitude of their CI coefficients is somewhat arbitrary and will lead to energies strongly depending on that choice. On the other hand, the inclusion of all configurations into the reference space would require a prohibitive amount of computer time. A further problem occurs with Th $7s^2\ 6d^1\ 7p^1\ ^3F$; since our atomic calculations have been carried out in symmetry D_{2h}, the $7p$ orbitals belong to the same irreducible representations (namely b_{1u}, b_{2u}, and b_{3u}) as some $5f$ orbitals. Some of the electronic states with $5f^1$ occupation, mentioned above, are of lower energy. Therefore, the calculation of high roots would have been required to obtain the energy of the given state. We decided to circumvent those problems by reduction of the active space.

The results are listed in Table IV and compared with experimental data,[28,29] obtained by $(2J+1)$-weighted averaging over term energies of the corresponding J levels of each LS state. Due to the relatively large number of states with similar energy a strong coupling between levels with the same J value and parity belonging to different LS states occurs. This mixing of states makes the comparison with experiment difficult. The $J=3/2$ levels of Th^+ $7s^2\ 6d^1\ ^2D$, $7s^1\ 6d^2\ ^4F$, and $6d^3\ ^4F$, for example, interact so strongly, that no level with a leading $7s^2\ 6d^1$ component is observed. Therefore the ground state of Th^+ is described best as $(7s+6d)^3$.[29] The experimental excitation energy between Th^+ $7s^2\ 6d^1\ ^2D$ and $7s^1\ 6d^2\ ^4F$ given in Table IV is the energy difference between the two lowest $J=3/2$ levels.[29]

For all states without occupied $5f$-orbitals, the agreement of our results with experimental data is very good. However, the results for Th $7s^2\ 6d^1\ 5f^1\ ^3H$ and Th^+ $7s^2$

TABLE III. Excitation and ionization energies of thorium from quasirelativistic (WB) and nonrelativistic (HF) finite-difference all-electron (AE) and pseudopotential calculations (PP). $\Delta E = E_{AE} - E_{PP}$, all values in eV.

	Configuration		E_{HFAE}	E_{HFPP}	ΔE_{HF}	E_{WBAE}	E_{WBPP}	ΔE_{WB}
	$6d^2 7s^2$	^3F	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	$6d^1 7s^2 7p^1$	^3F	3.7075	3.6974	0.0101	1.8603	1.8549	0.0053
	$7s^2 7p^2$	^3P	8.4033	8.3826	0.0208	4.7300	4.7230	0.0070
	$6d^3 7s^1$	^5F	−1.9343	−1.9200	−0.0143	−0.0464	−0.0427	−0.0038
	$6d^2 7s^1 7p^1$	^5G	0.2379	0.2451	−0.0072	0.8525	0.8407	0.0118
	$6d^1 7s^1 7p^2$	^5F	⋯	⋯	⋯	3.2486	3.2423	0.0063
	$5f^1 6d^1 7s^2$	^3H	−3.3979	−3.3506	−0.0473	1.3136	1.3143	−0.0007
	$5f^1 7s^2 7p^1$	^3G	−0.4183	−0.3734	−0.0449	2.9816	2.9941	−0.0125
	$5f^1 6d^2 7s^1$	^5I	−4.7132	−4.6569	−0.0563	1.6667	1.6722	−0.0055
	$5f^1 7s^1 7p^2$	^5G	0.0022	0.0579	−0.0557	⋯	⋯	⋯
	$5f^1 6d^3$	^5I	−4.3932	−4.3268	−0.0663	3.5787	3.5947	−0.0161
	$5f^2 7s^2$	^3H	−4.4550	−4.4185	−0.0365	4.8634	4.8497	0.0138
	$5f^2 6d^1 7s^1$	^5K	−5.1894	−5.1459	−0.0435	5.0488	5.0398	0.0090
	$5f^2 7s^1 7p^1$	^5I	−3.8480	−3.8002	−0.0478	6.1235	6.1245	−0.0010
	$5f^3 7s^1$	^5I	−2.9366	−2.9306	−0.0060	10.3840	10.3833	0.0007
1+	$6d^1 7s^2$	^2D	7.6820	7.6711	0.0109	5.7222	5.7315	−0.0093
1+	$7s^2 7p^1$	^2P	13.1801	13.1584	0.0217	9.2609	9.2649	−0.0040
1+	$6d^2 7s^1$	^4F	4.1489	4.1569	−0.0080	4.8756	4.8888	−0.0131
1+	$6d^1 7s^1 7p^1$	^4F	8.7408	8.7420	−0.0012	7.8807	7.8838	−0.0030
1+	$7s^1 7p^2$	^4P	14.2646	14.2574	0.0072	11.8064	11.8085	−0.0021
1+	$5f^1 7s^2$	^2F	3.2694	3.3111	−0.0418	6.4190	6.4365	−0.0175
1+	$5f^1 6d^1 7s^1$	^4H	0.6820	0.7380	−0.0561	6.0296	6.0460	−0.0163
1+	$5f^1 7s^1 7p^1$	^4G	4.5847	4.6435	−0.0588	8.8770	8.9091	−0.0320
1+	$5f^2 7s^1$	^4H	−0.4679	−0.4221	−0.0457	9.4152	9.4166	−0.0014
1+	$5f^3$	^4I	0.4812	0.4983	−0.0171	15.0382	15.0434	−0.0052

$5f^1\ ^2F$ show a somewhat larger error. We do not believe this to be a defect of the pseudopotential. A too high ionization and excitation energy has to be expected for states with f occupation due to our limited basis set and the errors in differential correlation energies related to them. While 8 f and 2 g functions are sufficient to correlate the $6d$ subshell and deliver a good description of the ground state, the 2 g functions alone are not able to correlate the $5f$ subshell equally well. The results should improve by increasing the number of g functions (or even including h functions). Further improvements should be obtained by correlating the $5d$ (and $5p$) subshell, but the computational effort will raise dramatically.

TABLE IV. Excitation and ionization energies of Th (in eV) obtained by LCAO self-consistent field (SCF-LCAO) calculations, configuration interaction calculations including single and double excitations [CI(SD)] with size-consistency correction [CI(SD)+SCC] and multireference averaged coupled-pair functional calculations (MRACPF), respectively.

		SCF-LCAO	CI(SD)	CI(SD)+SCC	MRACPF	Expt.	Ref.
$7s^2 6d^2$	^3F	0.00	0.00	0.00	0.00	0.00	28
$7s^1 6d^3$	^5F	−0.04	0.34	0.48	0.64	0.64	28
$7s^2 6d^1 5f^1$	^3H	1.37	1.79	1.83	1.49	1.05	28
$7s^2 6d^1 7p^1$	^3F	1.98	2.28	2.21	1.59	1.47	28
$1+7s^2 6d^1$	^2D	5.73	6.16	6.21	6.01	6.08	29
$1+7s^1 6d^2$	^4F	4.89	5.79	6.05	6.07	(6.14)	29
$1+7s^2 5f^1$	^2F	6.44	6.78	6.75	6.89	6.54	28
$1+7s^2 7p^1$	^2P	9.29	9.97	9.83	⋯	10.00	28

B. Molecular calculations on ThO

The results listed above show the reliability of our pseudopotentials and basis sets for use in atomic calculations. As a test for the transferability of our parameters to studies of molecules, calculations on some properties (namely the dissociation energy D_e, the vibrational wave number ω_e, and the bond length r_e) of the $^1\Sigma^+$ ground state of ThO have been carried out. For thorium we used our pseudopotentials (nonrelativistic as well as quasirelativistic) and the corresponding basis sets. For oxygen we applied a $(10s6p)/[5s3p]$ basis set,[30] augmented by 1 d function with an exponent of 1.2. To obtain the bond length r_e, we calculated the total energy for several Th–O distances, differing by 0.1 a.u. Afterwards, the minimum has been obtained by adjusting a third-degree polynomial to the four points of the lowest energy. The same polynomial has been used to calculate ω_e. In order to keep size consistency errors small, the dissociation energy D_e has been calculated by subtracting the total energies at $r=r_e$ and at $r=1000$ a.u.

Our calculations have been carried out in C_{2v} symmetry using the program system MOLPRO.[21–25] For each geometry we performed a SCF calculation first, then the resulting wave function was further optimized by a CASSCF. Following the work of Marian *et al.*[8] the active space was chosen to include 8 electrons distributed over 9 orbitals ($7s$ and $6d$ of thorium, $2p$ of oxygen) while the $2s$ orbital of oxygen was kept doubly occupied. This active space was taken as reference space for a subsequent internally contracted multireference CI calculation with all single and double excitations [MRCI(SI)].[24,25] Three series of calculations have been car-

TABLE V. Spectroscopic constants for the $^1\Sigma^+$ ground state of ThO.

	r_e (pm)	D_e (eV)	ω_e (cm^{-1})
Expt.[a]	184.0	9.00±0.09	896
Expt.[b]		8.87±0.15	
		8.79±0.13	
Marian *et al.*[c]	192.3	7.85	852
qrel. with f $(+2g)$			
SCF	182.9	6.07	943
SCF+CI(SD)	186.5	8.06	911
SCF+CI(SD)+SCC	187.4	8.20	891
SCF+MRCI(SD) $(+2g)$	185.2(183.7)	8.50(8.70)	905(925)
SCF+MRCI(SD)+SCC $(+2g)$	186.1(184.5)	8.67(8.87)	878(902)
CASSCF	188.2	8.92	876
CASSCF+MRCI(SD) $(+2g)$	188.4(185.7)	8.48(8.84)	870(879)
CASSCF+MRCI(SD)+SCC $(+2g)$	188.9(186.2)	8.64(8.87)	862(867)
qrel. without f			
SCF	196.1	3.71	869
CASSCF	202.5	7.02	944
CASSCF+MRCI(SD)	201.6	7.01	783
CASSCF+MRCI(SD)+SCC	202.0	7.20	772
nrel. with f			
SCF	171.7	4.66	1080
CASSCF	179.1	6.59	864
CASSCF+MRCI(SD)	183.2	7.58	834
CASSCF+MRCI(SD)+SCC	184.5	7.78	797

[a]Reference 36.
[b]Reference 32.
[c]Reference 8.

ried out; quasirelativistic calculations for comparison with experimental data, a corresponding series without f functions in the Th basis set to study the effect of the $5f$ orbitals on the chemical bonding in ThO, and finally a series of nonrelativistic calculations to obtain an idea about relativistic effects. The results are listed in Table V.

The spectroscopic constants derived from our quasirelativistic calculations including f functions are in reasonable agreement with experimental data. CASSCF+MRCI(SD) calculations without g functions yield a bond length being too long by 4.4 pm. Accounting for higher excitations by the use of the size consistency correction (SCC) of Davidson and Langhoff[27] elongates r_e slightly by 0.5 pm. The results are considerably improved by inclusion of 2 g functions. The bond length is shortened by 2.7 pm. The remaining errors of 1.7 and 2.2 pm (with SCC) are acceptable. For all calculations, the bond length is increased by 5–7 pm in the CASSCF, compared to SCF. The occurrence of a too long bond length as a result of CASSCF calculations may be attributed to the overestimation of the contributions of antibonding configurations to the wave function. Subsequent MRCI(SD) calculations may suffer from this unbalance of the reference space. In fact, single reference CI(SD) and MRCI(SD) calculations with SCF-orbitals lead to a further reduction of the bond length (1.7–3.2 pm). The derived values may even become better by including the O$(2s)$ and Th$(6p)$ orbitals into the active space, but this would considerably raise the computational effort. The effect of spin–orbit coupling on the bond length was studied in two-component SCF calculations[31] and found to be negligibly small.

Nevertheless, spin–orbit effects are important for the dissociation energy. Although they leave the molecular valence energy of a $^1\Sigma^+$ state virtually unchanged (decrease 0.03 eV), they lower the atomic ground states of Th $(6d^2 7s^2\ ^3H)$ and O $(2p^4\ ^3P)$ by 0.38 and 0.01 eV, respectively. Correcting our best CASSCF+MRCI(SD)+SCC result of 8.87 eV with these values and taking into account a zero point energy of approximately 0.06 eV leads to a binding energy of 8.45 eV, in good agreement with experimental values. As noted by Hildenbrand and Murad[32] the dissociation energy of 9 eV reported by Ackermann and Rauh[33] is expected to be an upper bound to the exact value, due to the choice of free energy functions. They report values of 8.86 and 8.79 eV, respectively.

Malli performed Dirac–Fock SCF calculations for five internuclear distances (3.077, 3.477, 3.877, 4.277, and 4.677 a.u.) of ThO.[34] We calculated the following properties from his data: $r_e=203.3$ pm, $D_e=0.52$ eV, and $\omega_e=939$ cm^{-1}. These results disagree dramatically with our quasirelativistic SCF data listed in Table V. The bond length is too long by 20.4 pm and the dissociation energy is too low by 5.55 eV (5.19 eV after correction for spin–orbit effects as mentioned above). The difference of the vibrational frequencies is very small (4 cm^{-1}), however, as shown below this results from a cancellation of errors. Malli applied a minimal basis set but did not include a $5f$ function for Th. This leads to a longer bond length, a smaller dissociation energy and a lower vibrational frequency, as shown by our SCF calculations using our quasirelativistic pseudopotential and our basis set without f-functions. These results show a somewhat better agreement with the data of Malli. The remaining differences are 6.2 pm for the bond length and 3.19 eV (2.83 eV) for the dissociation energy, respectively, only ω_e drops to 869 cm^{-1} increasing the difference to 70 cm^{-1}. Contracting our basis set to a

TABLE VI. Spectroscopic constants for the low-lying electronic states of ThO from spin–orbit CI calculations. Experimental data in parentheses and corresponding state labels were taken from Ref. 39; the term energy of the W state is an estimate from Ref. 40.

Ω	ΛS-states[a]	r_e (pm)	T_e (cm^{-1})	ω_e (cm^{-1})
0^+ (X)	99.5% $^1\Sigma^+$ 0.2% $^3\Pi$	182.5 (184.0)	0 (0)	948 (896)
1 (H)	98.9% $^3\Delta$ 0.6% $^3\Pi$ 0.1% $^1\Pi$	184.0 (185.8)	5 768 (5 317)	903 (857)
2 (Q)	93.3% $^3\Delta$ 5.3% $^1\Delta$ 0.8% $^3\Pi$	183.9 (185.6)	6 628 (6 128)	905 (858)
3 (W)	99.6% $^3\Delta$	183.7	8 213 (8 600)	908
0^-	91.0% $^3\Pi$ 8.3% $^3\Sigma^+$	184.4	11 051	899
0^+ (A)	95.8% $^3\Pi$ 0.1% $^1\Sigma^+$	184.3 (186.7)	11 328 (10 601)	901 (846)
2	88.5% $^1\Delta$ 6.1% $^3\Delta$ 4.0% $^3\Pi$	183.8	12 190	902
1 (B)	83.4% $^3\Pi$ 6.9% $^1\Pi$ 8.3% $^3\Sigma^+$ 0.5% $^3\Delta$	184.5 (186.4)	12 252 (11 129)	895 (843)
2	94.0% $^3\Pi$ 5.2% $^1\Delta$ 0.2% $^3\Delta$	184.2	14 504	900
1 (C)	67.9% $^1\Pi$ 14.6% $^3\Sigma^+$ 13.2% $^3\Pi$ 0.2% $^3\Delta$	185.7 (187.0)	17 156 (14 490)	874 (825)
0^-	91.2% $^3\Sigma^+$ 8.4% $^3\Pi$	185.3	18 884	875
1 (D)	76.0% $^3\Sigma^+$ 18.6% $^1\Pi$ 2.1% $^3\Pi$	185.4 (186.6)	19 040 (15 946)	873 (839)
0^+ (E)	52.3% $^1\Sigma^+$ 1.2% $^3\Pi$	185.9 (186.7)	19 084 (16 320)	855 (829)
2 (W)		186.2 (188.2)	19 401 (18 010)	858 (809)

[a]The percentages refer only to the ΛS-states arising from the Th^{2+} ($7s^2$,$7s^1 6d^1$) O^{2-} ($2p^6$) configurations and due to admixture of higher states do not add up to 100%, cf. text for the E and G states.

minimal basis including a $5f$ function on Th has the same effect on the bond length (190.8 pm) and the dissociation energy (3.35 eV), but raises the vibrational wave number (1087 cm^{-1}). Finally, elimination of the $5f$ functions from our minimal basis set leads to results in excellent agreement with those of Malli; we obtained r_e=202.2 pm, D_e=0.91 eV (0.55 eV after correction for spin–orbit effects), and ω_e=954 cm^{-1}. Therefore, we conclude that the deviations between our quasirelativistic SCF results and the Dirac–Fock SCF results of Malli may predominantly be attributed to the differences in the basis sets and not to a defect of our pseudopotential or quasirelativistic method.

The results presented here for the ThO ground state are, at the highest level of approximation, in somewhat better agreement with experiment than those of Marian *et al.*[8] This is mainly due both to the pseudopotential and the basis set. The pseudopotential used by Marian is derived following the frozen-core effective core potential ansatz of Pettersson, Gropen, and Wahlgren,[35] separating the wave function into an "inner core," an "outer core," and the valence shell. While the inner core ($1s-5s$,$2p-5p$,$3d-4d$) is replaced by the pseudopotential, the outer core ($6s$,$6p$,$5d$,$4f$) is represented by frozen orbitals expanded in the valence basis and only the valence shell ($7s$,$7p$,$6d$,$5f$) is treated variationally. As mentioned earlier, such a small valence shell will lead to large frozen-core errors (in the order of 1 eV for excitation energies between states with different $5f$ occupation). In addition the f basis used by Marian does not seem to be flexible enough. The smallest exponent is 0.45 (compared to about 0.06 in our basis set) and is not appropriate for describing the very diffuse $5f$ orbitals of Th. Considering the large f population of about 0.4 at equilibrium distance (Mulliken population analysis, $Th^{0.77+}7s^{0.73}7p^{0.10}6d^{0.02}5f^{0.38}O^{0.77-}2s^{1.95}2p^{4.80}3d^{0.02}$) a flexible f basis is indispensable. This fact is also emphasized by the results of our CASSCF+MRCI(SD)+SCC calculation without f functions; the bond length is elongated by 15 pm and the dissociation energy is lowered by 1.7 eV; this indicates significant participation of the $5f$ orbitals in bonding. The statement that thorium behaves more like a $6d$ rather than a $5f$ element based on the $7s^2 6d^2\ ^3H$ atomic ground state (in contrast to the Ce $6s^2 5d^1 4f^1\ ^1G$ atomic ground state) is only true as a first approximation, since polarization effects involving $5f$ orbitals are more important than for the d-transition elements.

As expected, the influence of f functions on the chemical bonding in ThO is even higher in the nonrelativistic case. The nonrelativistic $5f$ orbitals are considerably lower in energy, leading to a Th $5f^2 6d^1 7s^1\ ^5K$ atomic ground state at the SCF level. The f population in the $^1\Sigma^+$ state of ThO raises to about 0.7 and the bond length shortens by about 4 pm. Furthermore it has to be noted, that the $^1\Sigma^+$ state is not the ground state of nonrelativistic ThO. We suspect a nonrelativistic $\phi^1_{5f}\sigma^1_{6s}\ ^3\Phi$ ground state similar to that of CeO,[37] however, within this work no effort has been made to explore this issue further.

We also performed spin–orbit CI calculations for the low-lying electronic states of ThO using the programs developed by Pitzer.[38] In a simple model, which we apply for convenience for the following discussion, the ground state of ThO may be pictured as $Th^{2+}(7s^2)O^{2-}(2p^6)$. For the lowest-lying $^{1,3}\Sigma^+$, Π, Δ (Ω values 0^-, 0^+, 1, 2, 3) excited states the Th^{2+} ion is in a $7s^1 6d^1$ configuration. Besides these possibilities we also included the Th^{2+} $7s^1 5f^1$, $7s^1 7p^1$, and $6d^2$ configurations in the reference space. Configurations of Th^+O^- gave slightly higher energies and were excluded from the reference wave function. Single excitations for the eight valence electrons were allowed using 55 orbitals from a ground state SCF calculation with improved virtual orbitals for a $7s$ hole on Th. The results of these calculations together with the available experimental data[39] (recent references for higher excited states are given in Ref. 41; the estimate for the W state is given in Ref. 40) are summarized in Table VI. Due to the rather limited correlation treatment the bond lengths are too short by 2 pm and the vibrational frequencies are too high by typically 50 cm^{-1}. Up to a theoretical term energy of 15 000 cm^{-1} our calculated and the available experimental values agree within 10% (errors up to 0.15 eV). The trends in the experimental bond lengths and vibrational frequencies as well as the observed energetic ordering of the spin–orbit components arising mainly from the $^1\Sigma^+$, $^3\Delta$, and $^3\Pi$ states of the $Th^{2+}(7s^2)O^{2-}(2p^6)$ and $Th^{2+}(7s^16d^1)O^{2-}(2p^6)$ configurations are well reproduced by our calculations. In addition to the experimentally observed $\Omega=0^+$ (X), 1 (H), 2 (Q), 3 (W), 0^+ (A), and 1 (B) states we found one $\Omega=0^-$ and two $\Omega=2$ states below 15 000 cm^{-1}. The $\Omega=0^-$ and the higher $\Omega=2$ state arise mainly from the $^3\Pi$ state, whereas the lower $\Omega=2$ state is due to the $^1\Delta$ state. The agreement for the states with theoretical term energies of 15 000–20 000 cm^{-1} is only within 20% (errors up to 0.40 eV), which may be due to the absence of reference configurations corresponding to Th^+O^- as well as missing dynamic electron correlation. The experimentally observed $\Omega=1$ (C) and 1 (D) states as well as the theoretically found $\Omega=0^-$ state arising mainly from $^3\Sigma^+$ are still due to a $Th^{2+}(7s^16d^1)O^{2-}(2p^6)$ configuration, whereas the experimentally observed $\Omega=0^+$ (E) state is essentially a mixture of 52.3% Th^{2+} $(7s^16d^1)$ O^{2-} $(2p^6)$ $^1\Sigma^+$ and 40.1% Th^{2+} $(6d^2)$ O^{2-} $(2p^6)$ $^3\Sigma^-$ according to our calculations. The $\Omega=2$ (G) state found in experiment correspond to 92.4% to the lowest component of a $^3\Phi$ state arising from a $Th^{2+}(6d^2)O^{2-}(2p^6)$ configuration. In view of the limited accuracy of the current calculations we restrict the systematic analysis of the electronic states of ThO to those with theoretical term energies below 20 000 cm^{-1}. Further investigations including more accurately correlation effects have to be carried out in order to assign unambiguously the electronic states of ThO observed above 15 000 cm^{-1}. In agreement with Marian *et al.* we did not find a $^3\Phi$ state resulting from a Th^{2+} $(7s^15f^1)$ O^{2-} $(2p^6)$ configuration at low energies, its lowest $\Omega=2$ component ($r_e=186.4$ pm, omega$_e=812$ cm^{-1}) having a term energy of 32 198 cm^{-1}. We note that CeO has a Ce^{2+} $(6s^14f^1)$ O^{2-} $(2p^6)\Omega=2(^3\Phi)$ ground state, reflecting the much stronger relativistic destabilisation of the $5f$ orbitals in Th compared to the $4f$ orbitals in Ce.

IV. CONCLUSION

Accurate energy-adjusted, nonrelativistic as well as (one and two-component) quasirelativistic pseudopotentials for the $5f$ elements actinium to lawrencium have been derived. Energy-optimized $(12s11p10d8f)/[8s7p6d4f]$ valence basis sets have been presented. The deviations between atomic excitation and ionization energies obtained by quasirelativistic finite difference all-electron SCF calculations and corresponding values obtained by use of our pseudopotentials and basis sets, are less than 0.2 eV. Results of test calculations on Th and ThO, including electron correlation and spin–orbit effects, showed reasonable agreement with experimental data. The derived pseudopotentials and basis sets should be a reliable tool for accurate quantum chemical calculations on small actinide compounds.

ACKNOWLEDGMENTS

Thanks are due to Professors W. Nieuwpoort, R. M. Pitzer, W. H. E. Schwarz, and H.-J. Werner for providing their programs. The financial support of the Deutsche Forschungsgemeinschaft (DFG) to one of us (M. D.) is gratefully acknowledged. Part of this work was performed using a grant of computer time from the Ohio Supercomputer Center.

[1] M. Krauss and W. J. Stevens, Annu. Rev. Phys. Chem. **35**, 357 (1984).
[2] P. A. Christiansen, W. C. Ermler, and K. S. Pitzer, Annu. Rev. Phys. Chem. **36**, 407 (1985).
[3] U. Häussermann, M. Dolg, H. Stoll, H. Preuss, P. Schwerdtfeger, and R. M. Pitzer, Mol. Phys. **78**, 1211 (1993).
[4] G. B. Bachelet, D. R. Hamann, and M. Schlüter, Phys. Rev. B **26**, 4199 (1982).
[5] W. R. Wadt, J. Am. Chem. Soc. **103**, 6053 (1981).
[6] P. J. Hay, J. Chem. Phys. **79**, 5469 (1983).
[7] W. R. Wadt, J. Chem. Phys. **86**, 339 (1987).
[8] C. M. Marian, U. Wahlgren, O. Gropen, and P. Pyykkö, J. Mol. Struct. (Theochem) **169**, 339 (1988).
[9] W. C. Ermler, R. B. Ross, and P. A. Christiansen, Int. J. Quantum Chem. **40**, 829 (1991).
[10] M. Dolg, H. Stoll, A. Savin, and H. Preuss, Theor. Chim. Acta **75**, 173 (1989).
[11] M. Dolg, H. Stoll, and H. Preuss, J. Chem. Phys. **90**, 1730 (1989).
[12] Program MCHF77, C. Froese Fischer, Pennsylvania State University, 1977, modified by M. Dolg, 1987.
[13] C. Froese Fischer, *The Hartree–Fock Method for Atoms: A Numerical Approach* (Wiley, New York, 1976).
[14] Program LSTERMS, J. T. Van Montford, H. U. Van Piggelen, G. Aissing, W. C. Nieuwpoort, Groningen, 1983.
[15] J. H. Wood and A. M. Boring, Phys. Rev. B **18**, 2701 (1978).
[16] R. D. Cowan and D. C. Griffin, J. Opt. Soc. Am. **66**, 1010 (1976).
[17] J. C. Slater, Phys. Rev. **81**, 385 (1951).
[18] J. P. Perdew and A. Zunger, Phys. Rev. B **23**, 5048 (1981).
[19] R. M. Pitzer and N. W. Winter, J. Phys. Chem. **92**, 3061 (1988).
[20] A file with pseudopotential parameters and corresponding valence basis sets for almost all elements of the Periodic Table is available by electronic mail. Requests should be addressed to cfag1019@servus.rus.uni-stuttgart.de.
[21] MOLPRO is a package of *ab initio* programs written by H.-J. Werner and P. J. Knowles with contributions of J. Almlöf, R. Amos, S. Elbert, K. Hampel, W. Meyer, K. Peterson, R. Pitzer, and A. Stone.
[22] H.-J. Werner and P. J. Knowles, J. Chem. Phys. **82**, 5053 (1985).
[23] P. J. Knowles and H.-J. Werner, Chem. Phys. Lett. **115**, 259 (1985).
[24] H.-J. Werner and P. J. Knowles, J. Chem. Phys. **89**, 5803 (1988).
[25] P. J. Knowles and H.-J. Werner, Chem. Phys. Lett. **145**, 514 (1988).
[26] H.-J. Werner and P. J. Knowles, Theor. Chim. Acta **78**, 175 (1990).
[27] S. R. Langhoff and E. R. Davidson, Int. J. Quantum Chem. **8**, 61 (1974).
[28] J. Blaise and J.-F. Wyart, Energy levels and atomic spectra of actinides, in International Tables of selected constants 20. Tables Internationales de constantes, Paris, 1992.
[29] W. C. Martin, L. Hagan, J. Reader, and J. Sugar, J. Phys. Chem. Ref. Data **3**, 771 (1974).
[30] T. H. Dunning, J. Chem. Phys. **55**, 716 (1971).
[31] UHREL is a two-component relativistic SCF program written by P. Hafner and W. H. E. Schwarz, with contributions from M. Dolg and P. Schwerdtfeger; (a) R. Ahlrichs, Theor. Chim. Acta. **33**, 157 (1974); (b) P. Hafner and W. H. E. Schwarz, Chem. Phys. Lett. **65**, 357 (1979).
[32] D. L. Hildenbrand and E. Murad, J. Chem. Phys. **61**, 1232 (1974).
[33] R. J. Ackermann and E. G. Rauh, High Temp. Sci. **5**, 463 (1973).
[34] G. L. Malli, in R. Salahub and M. Zerner, ACS Symp. Ser. **394**, 291 (1989).
[35] L. G. M. Pettersson, U. Wahlgren, and O. Gropen, Chem. Phys. **80**, 7 (1983).
[36] K. P. Huber and G. Herzberg, *Constants of Diatomic Molecules* (Van Nostrand Reinhold, New York, 1979).
[37] M. Dolg, H. Stoll, and H. Preuss, J. Mol. Struct. (Theochem) **231**, 243 (1991).
[38] Programs ARGOS, CNVRT, SCFPQ, LSTRN, CIGNLS, CIDG, SCF, and spin–orbit CI program, A. Chang, R. M. Pitzer, J. Am. Chem. Soc. **111**, 2500 (1989).
[39] G. Edvinsson and A. Lagerqvist, J. Mol. Spectrosc. **113**, 93 (1985).
[40] G. Edvinsson and A. Lagerqvist, J. Mol. Spectrosc. **122**, 428 (1987).
[41] G. Edvinsson and A. Lagerqvist, Phys. Scr. **41**, 316 (1990).